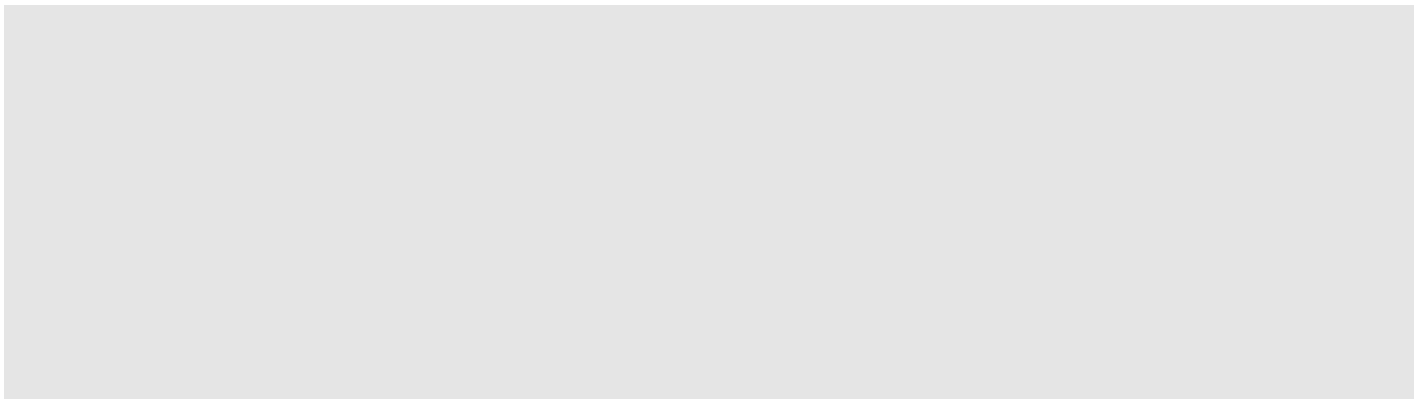


Speak Everything and (let them) Rise!



Dennie Declercq // 10-02-2022 // ASEAN Cloud Community

Intro



Content

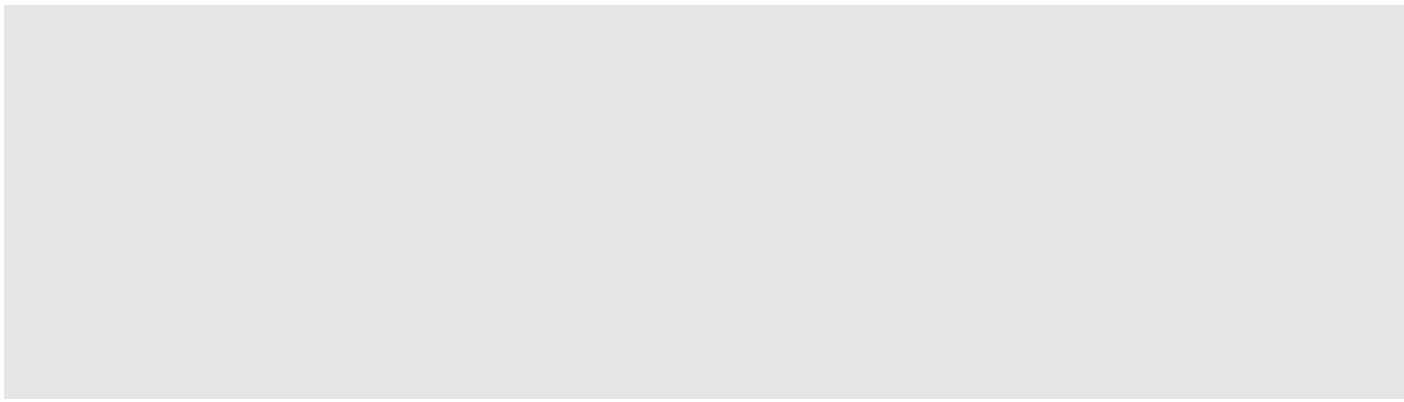
- Intro
- A primer into Accessibility
- Speech for Accessibility
- Microsoft Cognitive Speech
- Speech for WPF
- Speech for Web
- Thanks // Q&A

Who am I?



- Dennie Declercq
- President/ Developer
DDSoft
- Microsoft MVP
- dennie@ddsoft.be
- @DennieDeclercq
- #DreamingIsBelieving

A primer into Accessibility



People with disabilities



- Learning Disability
 - Reading
 - Understanding / Cognition
 - Structure
- Autism / Autism Spectrum Disorder
 - Structure
 - Confirmation
- Dyslexia
 - Reading
- Mental Health
 - Structure (sometimes)
 - Confirmation

Learning Disability?

Intellectual Functioning

Also called intelligence

Mental capacity

IQ score between 70-75

Adaptive Behavior

Conceptual Skills

- Language/literacy
- Money
- Time/numbers

Social Skills

- Social responsibility
- Self-esteem

Practical skills

- Activities daily living
- Personal care / healthcare

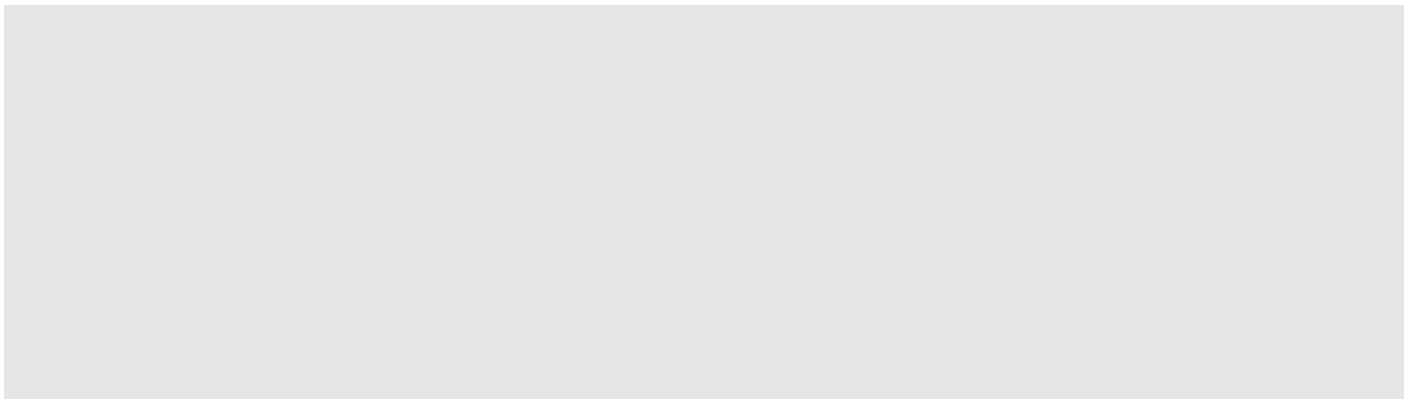
Oh yes...

- And the well known Accessibility too
 - Blindness/ Vision loss
 - Auditory disability
 - Motoric restrictions

The Persona Spectrum

	Permanent	Temporary	Situational
Touch	One arm	Arm injury	New parent
See	Blind	Cataract	Distracted driver
Hear	Deaf	Ear infection	Bartender
Speak	Non-verbal	Laryngitis	Heavy Accent

Speech for Accessibility



What is computer speech?

- Official name of technology: **Speech Synthesis**
- **Speech synthesis** is the artificial production of human [speech](#). A computer system used for this purpose is called a **speech computer** or **speech synthesizer**, and can be implemented in [software](#) or [hardware](#) products. A **text-to-speech (TTS)** system converts normal language text into speech; other systems render [symbolic linguistic representations](#) like [phonetic transcriptions](#) into speech.^[1] The reverse process is [speech recognition](#).
- Source: Wikipedia: <https://go.ddsoft.be/ttswiki>

A brief history of computer speech

- The [Mattel Intellivision](#) game console offered the [Intellivoice](#) Voice Synthesis module in 1982.
- Also released in 1982, [Software Automatic Mouth](#) was the first commercial all-software voice synthesis program.
- Arguably, the first speech system integrated into an [operating system](#) was the 1400XL/1450XL personal computers designed by [Atari, Inc.](#) using the Votrax SC01 chip in 1983.
- The first speech system integrated into an [operating system](#) that shipped in quantity was [Apple Computer's MacInTalk](#). The software was licensed from third-party developers Joseph Katz and Mark Barton (later, SoftVoice, Inc.) and was featured during the 1984 introduction of the Macintosh computer
- Android 1.6 has Text to Speech possibility (2009)
- Used in [Alexa](#) and as [Software as a Service](#) in AWS[58] (from 2017).

I am Belgian

Lernout &
Hauspie



Speech vs Read Aloud/ Voice Over

Read Aloud/ Voice Over

Blind/ Vision Loss



How does it work?

➔ Reads sequentially whole screen

(Separate) Speech

Learning Disability

Dyslexia



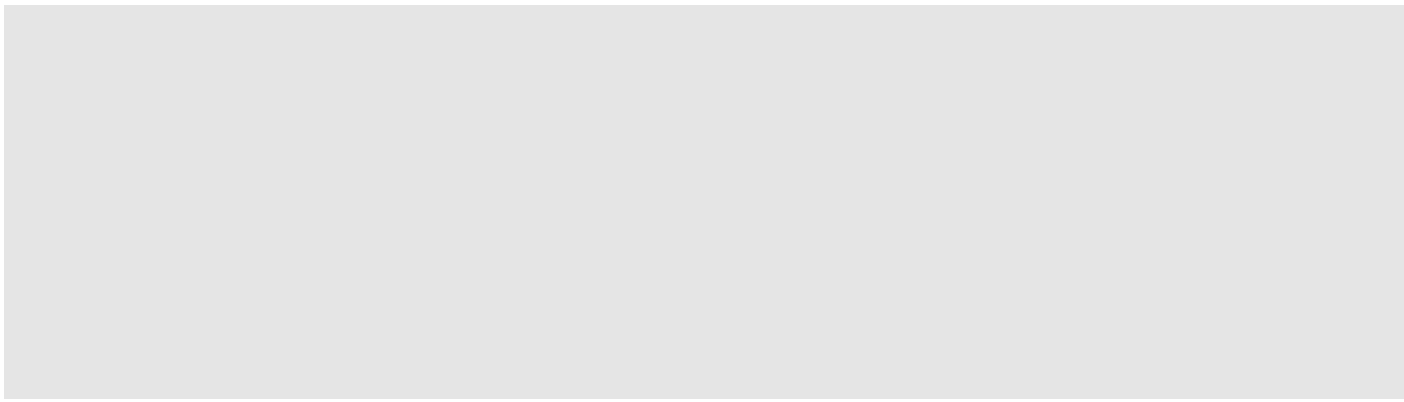
How does it work?

➔ Click a control and read aloud

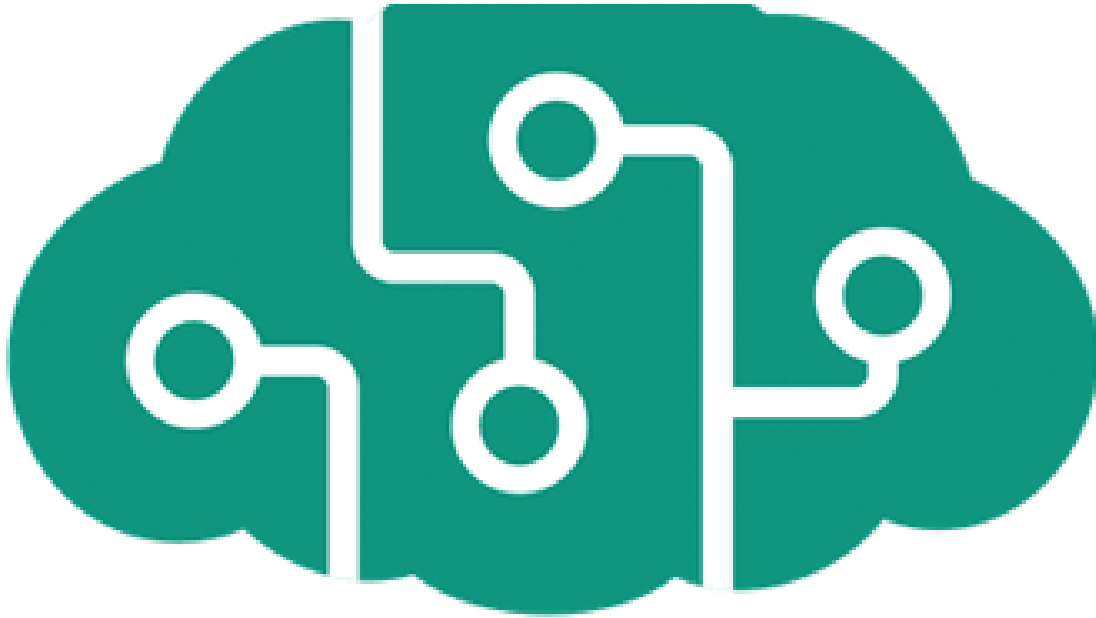
(Separate) Speech use cases

- Dyslexia
- Learning Disabilities
- The need for confirmations
- (Public) transport

Microsoft Cognitive Speech



A part of Microsoft Cognitive Services



- Microsoft “developer” AI solution.
- Part of Azure
- Developer focused
- No need to be a math rockstar
- No need to be a science engineer

Making a Speech account

[Home](#) > [New](#) > [Marketplace](#) >

Speech 
Microsoft



Speech  [Save for later](#)
Microsoft

Create

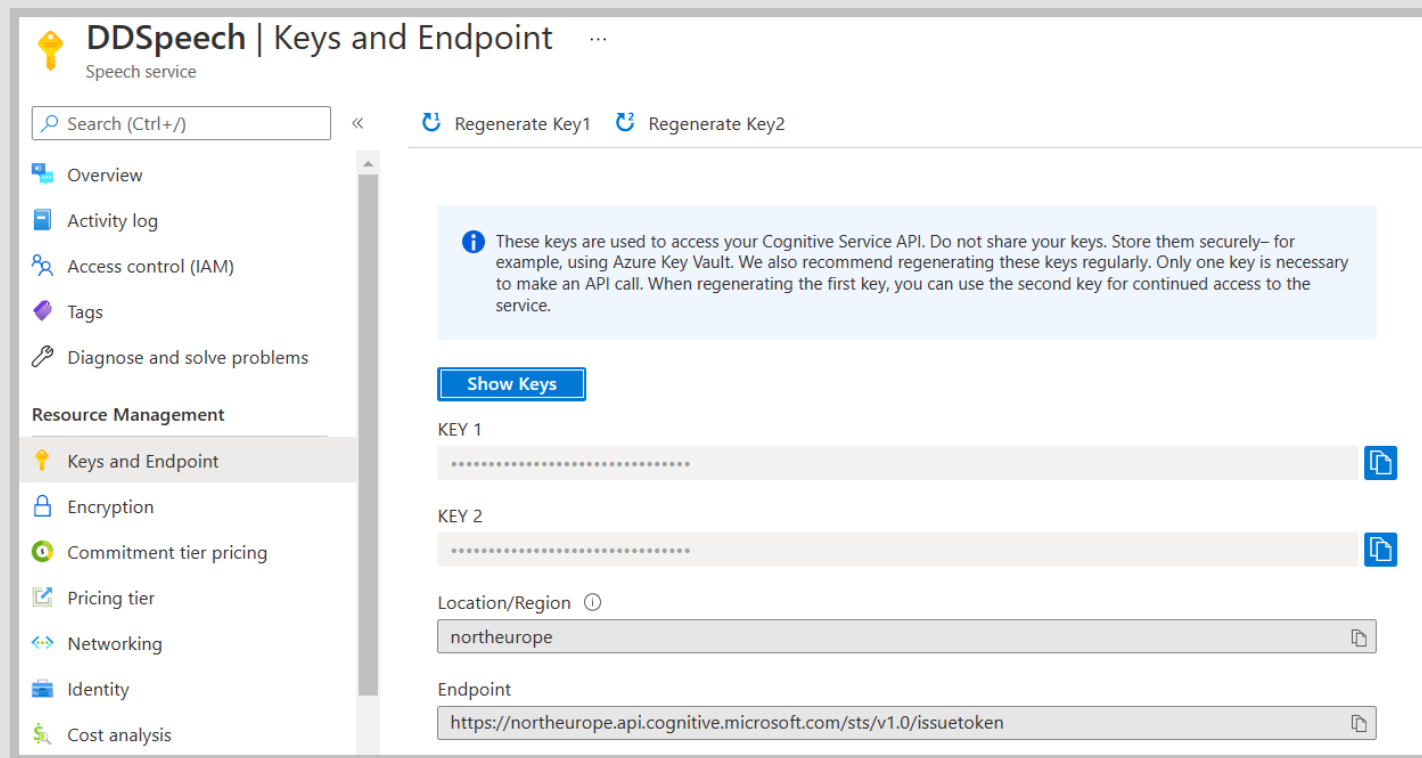
Overview

[Plans](#)

[Usage Information + Support](#)

Transcribe audible speech into readable, searchable text. Add real-time speech-enabled apps and services using the programming languages you a

Getting your keys & endpoints



The screenshot shows the 'DDSpeech | Keys and Endpoint' page in the Azure portal. The left sidebar contains a search bar and a list of navigation items: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, and a 'Resource Management' section with links to Keys and Endpoint (selected), Encryption, Commitment tier pricing, Pricing tier, Networking, Identity, and Cost analysis. The main content area has a title bar with 'DDSpeech | Keys and Endpoint' and a 'Speech service' subtitle. Below the title bar is a search bar and two 'Regenerate' buttons. A blue information box contains a warning about key security. A 'Show Keys' button is present. Below it, 'KEY 1' and 'KEY 2' are displayed as masked strings with copy icons. The 'Location/Region' is set to 'northeurope' with a dropdown icon. The 'Endpoint' is shown as 'https://northeurope.api.cognitive.microsoft.com/sts/v1.0/issuetoken' with a copy icon.

DDSpeech | Keys and Endpoint ...
Speech service

Search (Ctrl+/,) << Regenerate Key1 Regenerate Key2

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems

Resource Management

- Keys and Endpoint
- Encryption
- Commitment tier pricing
- Pricing tier
- Networking
- Identity
- Cost analysis

These keys are used to access your Cognitive Service API. Do not share your keys. Store them securely– for example, using Azure Key Vault. We also recommend regenerating these keys regularly. Only one key is necessary to make an API call. When regenerating the first key, you can use the second key for continued access to the service.

Show Keys

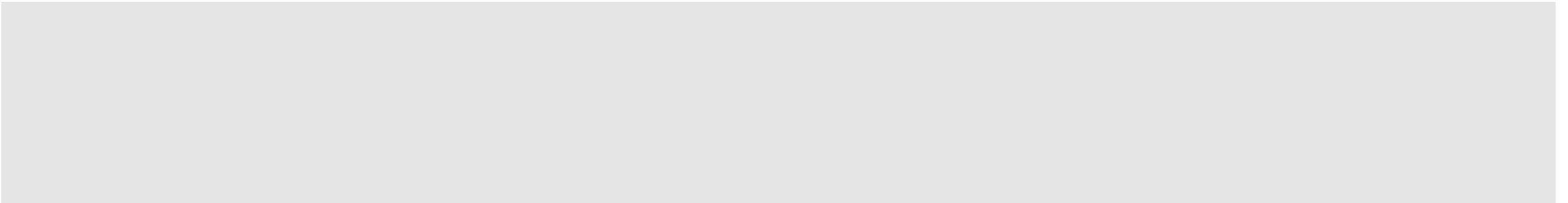
KEY 1
.....

KEY 2
.....

Location/Region ⓘ
northeurope

Endpoint
https://northeurope.api.cognitive.microsoft.com/sts/v1.0/issuetoken

DEMO: Making a Speech Service and accessing credentials



NUGET Package Speech SDK

Browse

Installed

Updates

Search (Ctrl+L)



Include prere



Microsoft.CognitiveServices.Speech by Microsoft

Microsoft Cognitive Services Speech SDK

Microsoft.CognitiveServices.Speech
Microsoft Cognitive Services Speech SDK

Code to speak (SDK)

```
using Microsoft.CognitiveServices.Speech;  
var _config = SpeechConfig.FromSubscription(KEY,REGION);  
using (SpeechSynthesizer synth = new SpeechSynthesizer(_config))  
{  
    await synth.SpeakTextAsync(text);  
}
```

REST: Making AccessToken

```
public async Task SetAuthorizationTokenAsync(string fetchUri, string subscriptionKey)
{
    HttpClient client = new HttpClient();
    client.DefaultRequestHeaders.Add("Ocp-Apim-Subscription-Key", subscriptionKey);
    var response = await
client.PostAsync("https://northeurope.api.cognitive.microsoft.com/sts/v1.0/issueToken", null);
    _speech.AccessToken = await response.Content.ReadAsStringAsync();
}
```

Accessing REST Speech Service

RESULT = STREAM GetTTS

```
string body = "<speake version='1.0' xml:lang='en-US'><voice xml:lang=''" + _speech.Language + "' xml:gender='Male'  
name = 'en-US-ChristopherNeural'>" + text + "</voice></speake>";
```

```
HttpClient client = new HttpClient();
```

```
HttpContent content = new StringContent(body);
```

```
content.Headers.ContentType = new  
System.Net.Http.Headers.MediaTypeHeaderValue("application/ssml+xml");
```

```
client.DefaultRequestHeaders.Add("X-Microsoft-OutputFormat", "riff-16khz-16bit-mono-pcm");
```

```
client.DefaultRequestHeaders.Authorization = new  
System.Net.Http.Headers.AuthenticationHeaderValue("Bearer", accessToken);
```

```
client.DefaultRequestHeaders.Add("User-Agent", "TextHelp");
```

```
HttpResponseMessage response = await client.PostAsync(_endpoint, content);
```


Saving AudioStream to .mp3 file

```
var readStream = await GetTTS(_speech.AccessToken, text);
```

```
FileStream writeStream = File.Create(path);
```

```
int Length = 256;
```

```
Byte[] buffer = new Byte[Length];
```

```
int bytestoRead = readStream.Read(buffer, 0, Length);
```

```
while (bytestoRead > 0)
```

```
{
```

```
    writeStream.Write(buffer, 0, bytestoRead);
```

```
    bytestoRead = readStream.Read(buffer, 0, Length);
```

```
}
```

```
readStream.Close();
```

```
writeStream.Close();
```

API style vs SDK style

API

- Save audio to file
- Need to develop the file generation code
- Everywhere possible
- NO Send audio directly to speakers

SDK

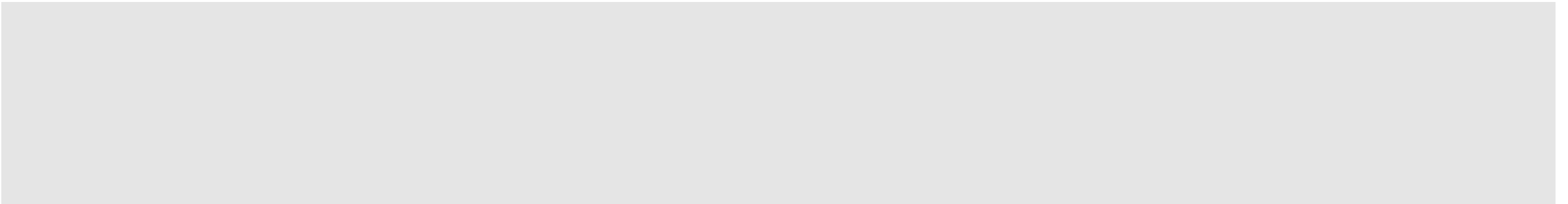
- Send audio directly to Speakers
- Save audio to file
- Easy to use
- Few lines of code
- Not everywhere possible

Things to keep in mind

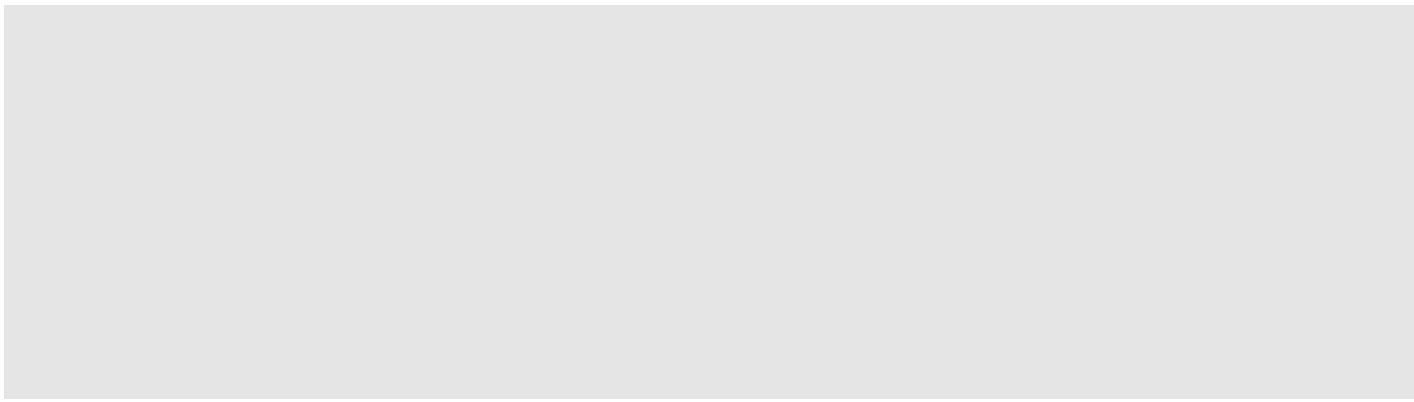
- Character Count
- Pause & more
- The pay model is via used characters

DEMO: SPEECH via SDK

DEMO: SPEECH via web API



Speech for WPF

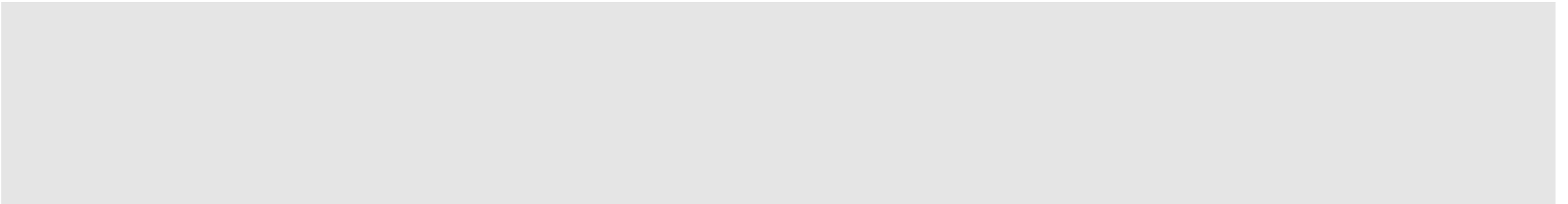


Accessing The Speech SDK thru WPF

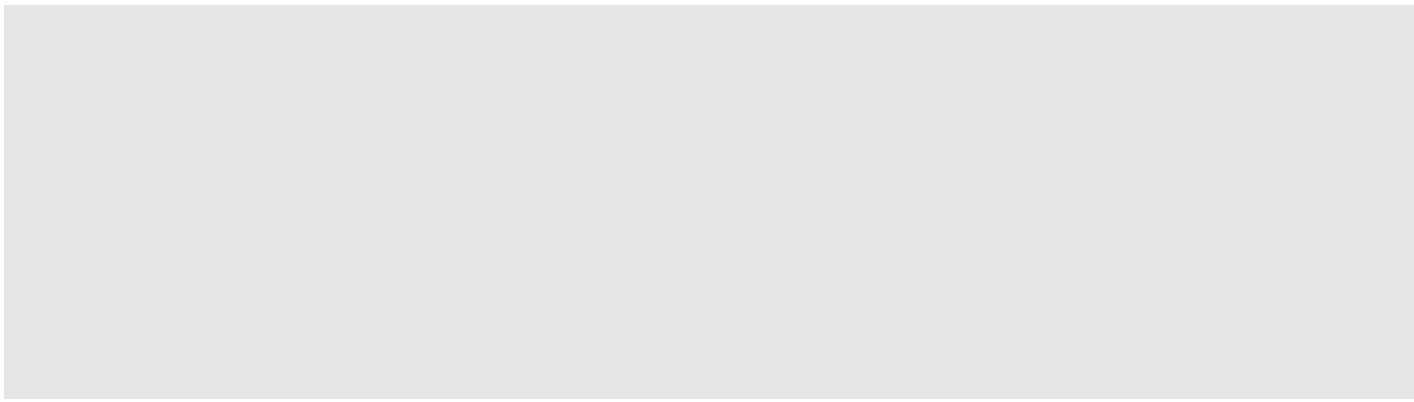


- Why?
 - Make windows applications accessible for people with learning disabilities.
 - Use a read function on fields for to check written tekst for people with dyslexia
- What
 - Each control is possible
 - The help page
 - Nessecary elements

DEMO: SPEECH for WPF



Speech for Web



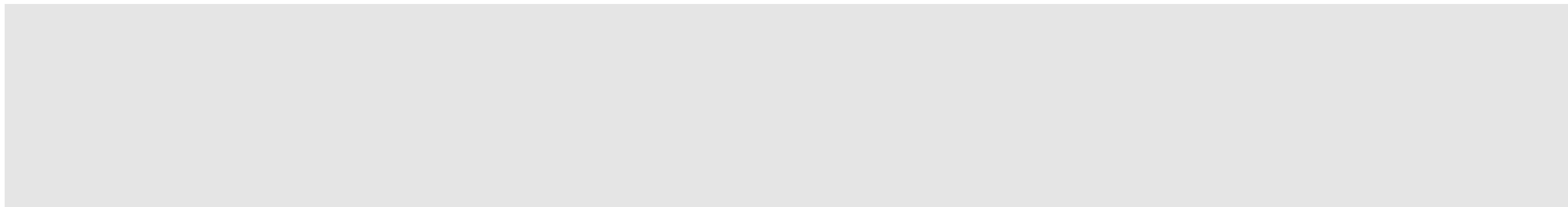
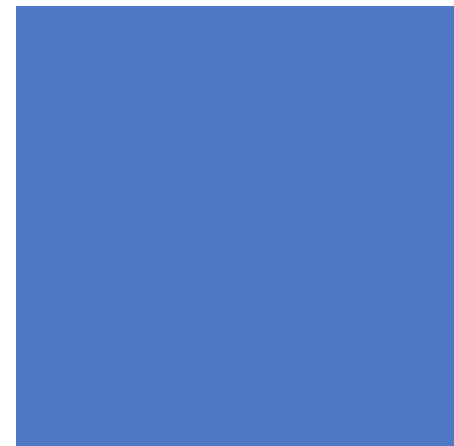
Speech on Hosted Web Apps

- SDK Speech can be added on ASP.NET hosted websites.
- This provides a whole new way of web accessibility for people with learning disabilities.
- This is the future if we want to include everyone.

Speech on WASM

- On Blazor running on Web Assembly you can only use REST Speech.
- Generate and save audio files to:
 - Web project (Azure Static Websites)
 - Azure Blob Storage

DEMO: Speech for Web



Thanks // Q&A



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